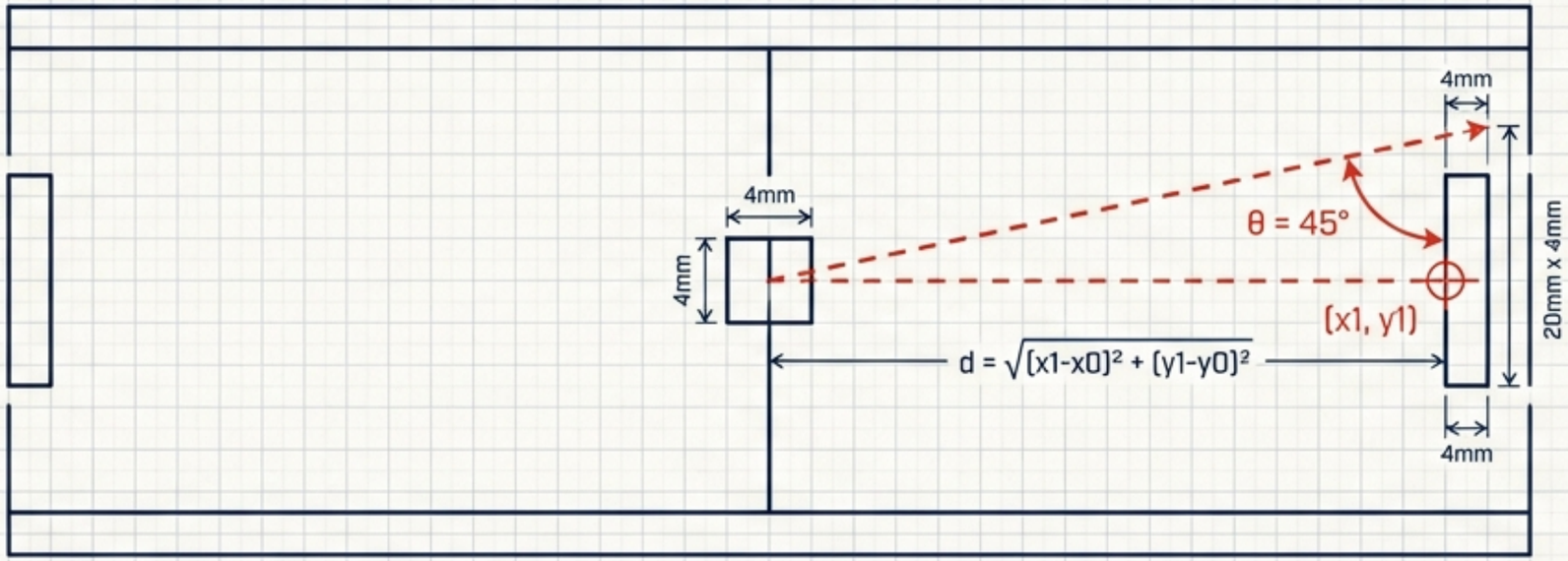
SCALE mm | REV. 3

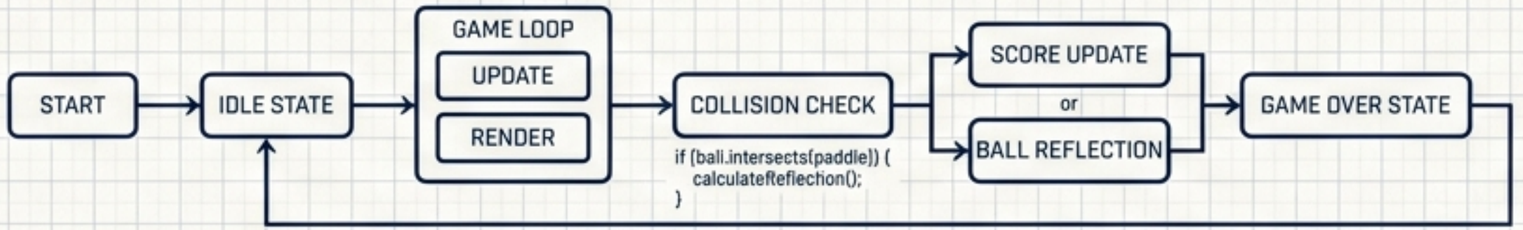
Week 1: Pong & The Art of Domain Modeling

50 Weeks GameDev Challenge

Do not build a screen. Build a system.



DOMAIN MODEL: GAME STATE



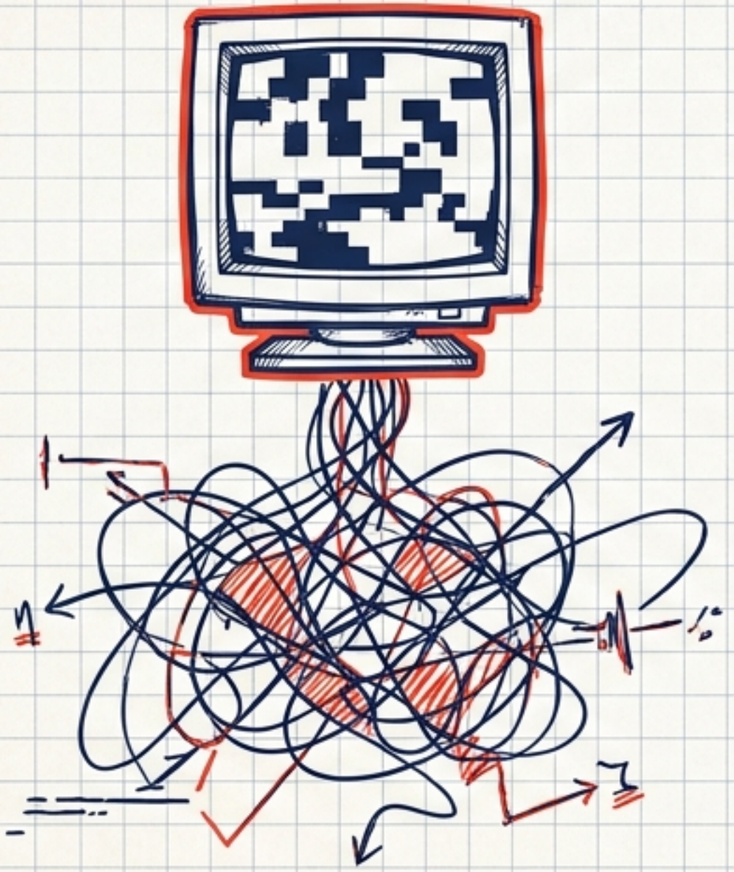
SCALE: 2.5 mm | REV. 3

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THE TRAP OF THE SCREEN

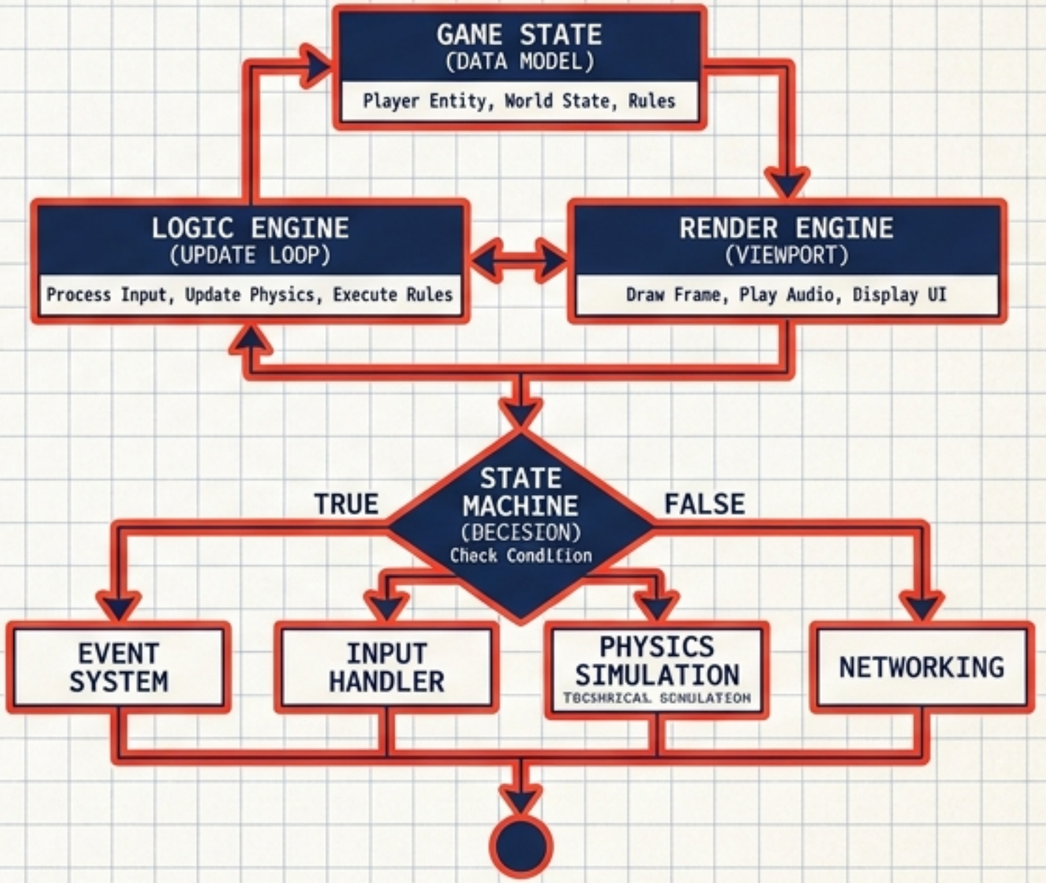
THE ILLUSION (AMATEUR MINDSET)

CONCEPT: Games are moving pictures.
PROCESS: Start with rendering.
RESULT: Brittle, tangled code.



THE REALITY (ARCHITECT MINDSET)

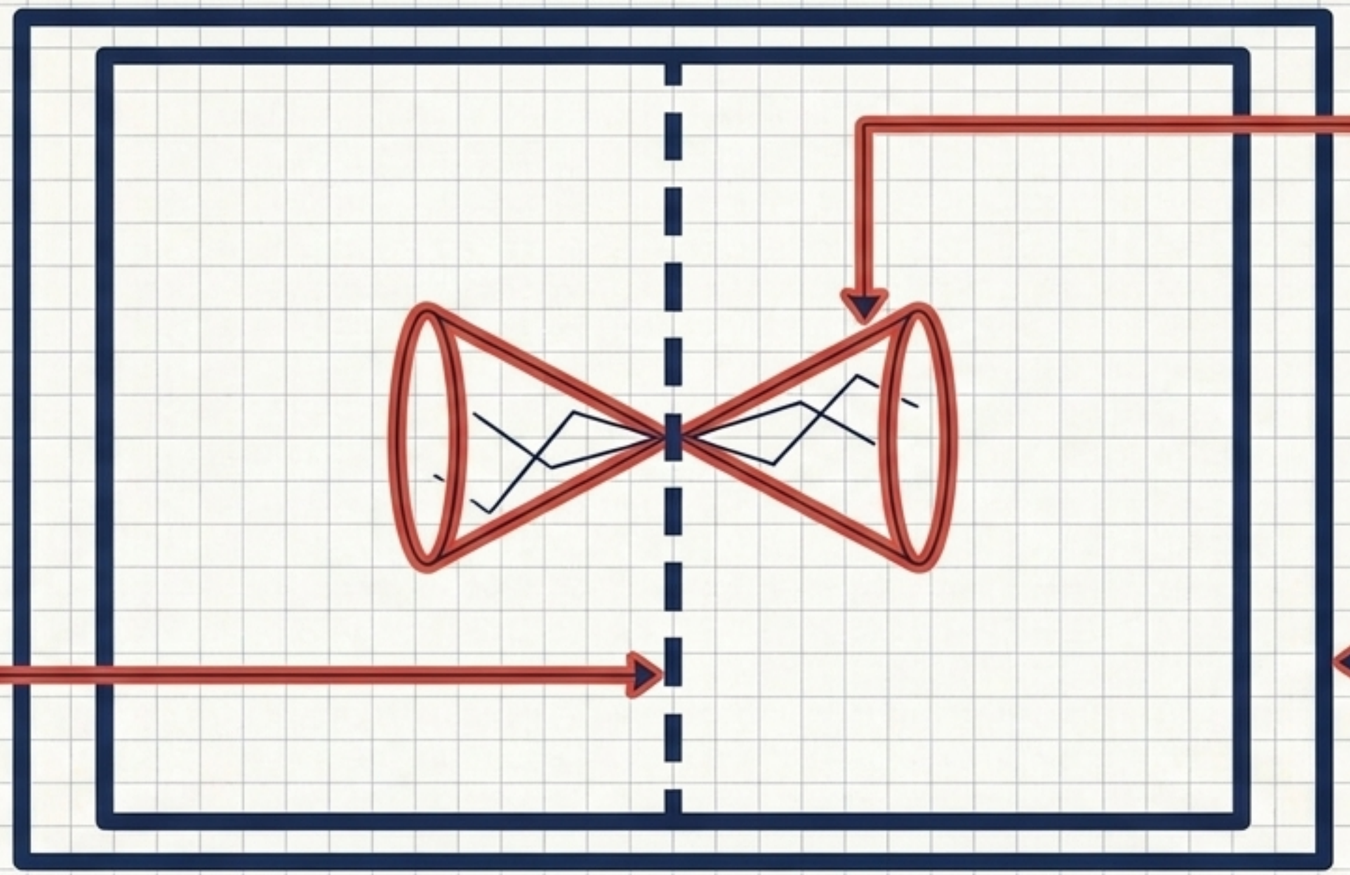
CONCEPT: Interactive state machines.
PROCESS: Domain modeling first.
RESULT: Clean, scalable architecture.



NOTE: DIAGRAMATIC REPRESENTATION DOES NOT IMPLY INPLEMENTATION DETAIL. CONCEPTUAL MODELING ONLY. DWG. NO. 042-A.

Gemini Notebook

STEP 0: THE DOMAIN OF PONG



INITIALIZATION: CENTER-SPAWN ONLY. RANDOM FIRE WITHIN DEFINED TRAJECTORY CONES.

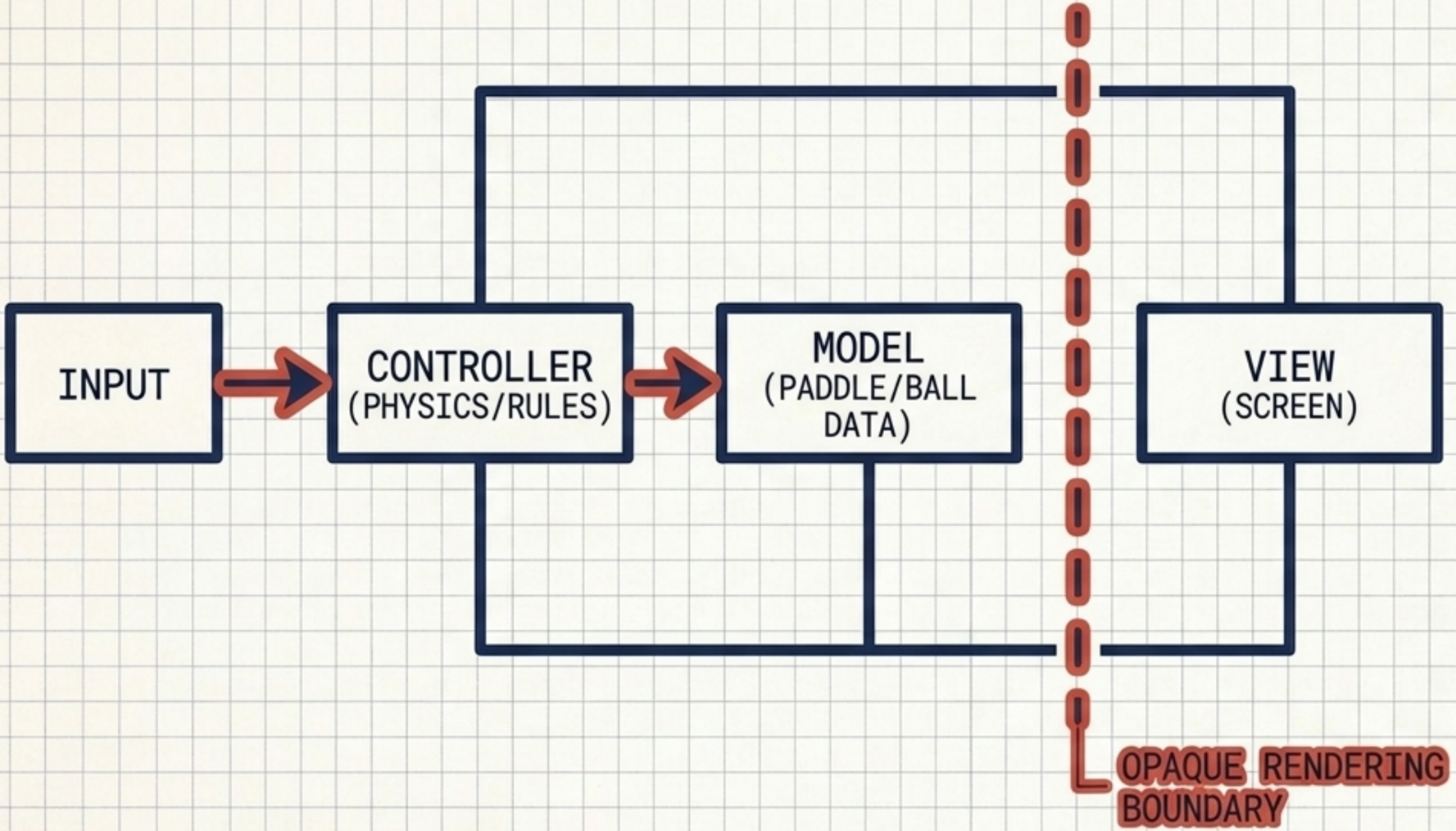
SYMMETRY: PHYSICS MIRROR PERFECTLY THE Y-AXIS. IDENTICAL PLAY FOR P1 AND P2.

THE OBJECTIVE: PENETRATE OPPONENT'S SPACE. STRICT WIN CONDITION: FIRST TO 4.

DWG. NO. 042-B

Gemini Notebook

ARCHITECTURE: THE BLIND ENGINE



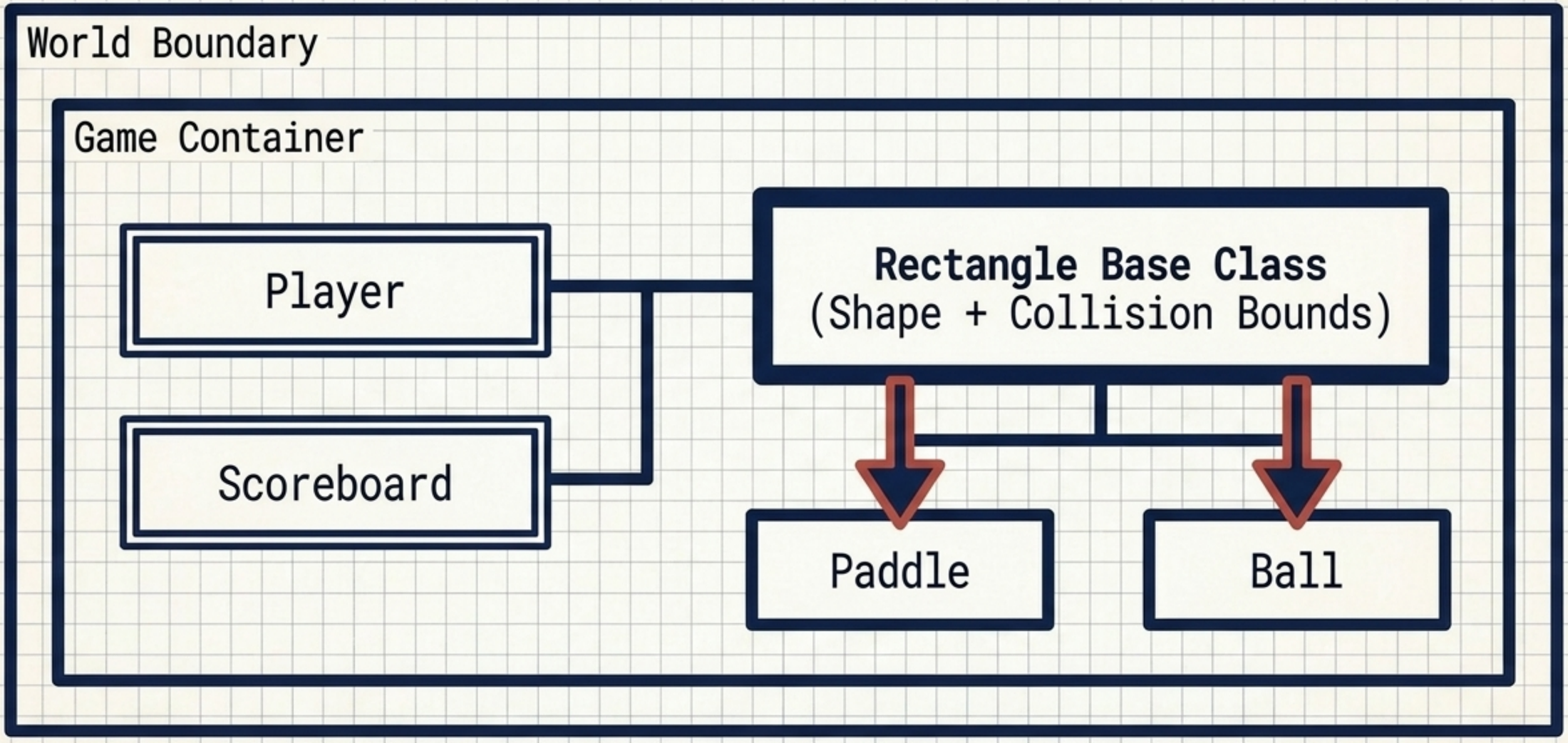
- **SEPARATE GAME LOGIC:** The engine must never care how pixels are pushed.
- **MODEL FIRST:** Design entities as independent mathematical constructs.
- **OPAQUE DRAW CALLS:** The logic crunches math, shouts "Draw!", and remains blind.

NOTE: DIAGRAMATIC REPRESENTATION DOES NOT IMPLY IMPLEMENTATION DETAIL. CONCEPTUAL MODELING ONLY.

DWG. NO. 842-C

Gemini Notebook

THE CAST OF ENTITIES



Game Container: Master orchestrator of all states.

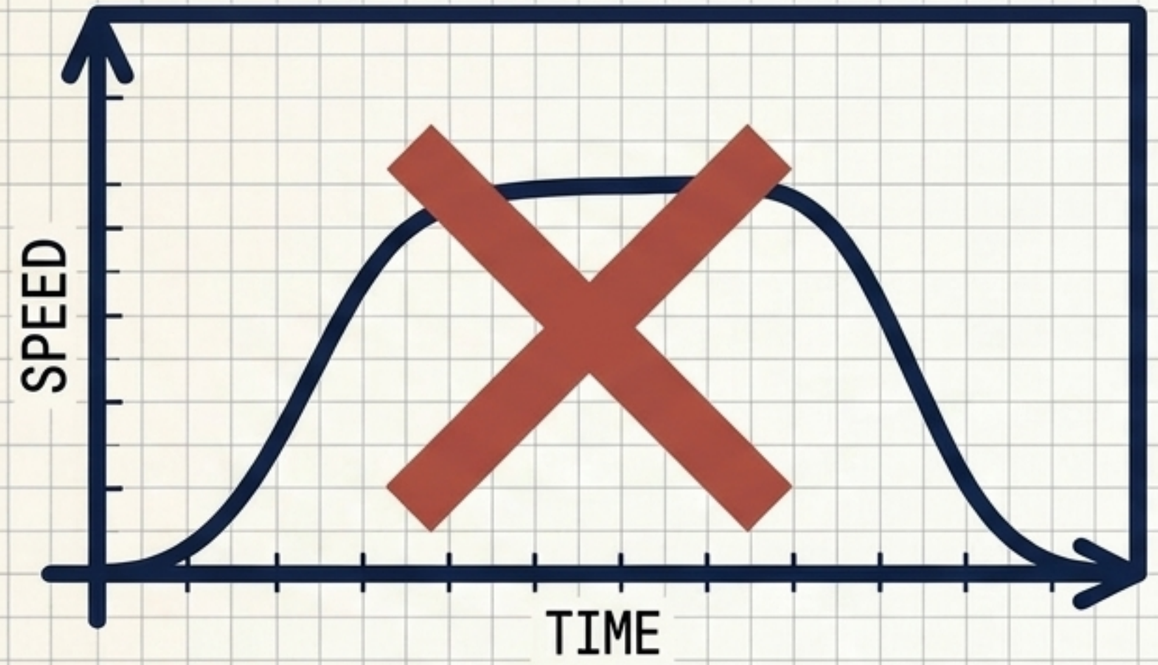
The Primitive: Every physical actor inherits from Rectangle.

The Advantage: Inherit collision properties for free on every spawned object.

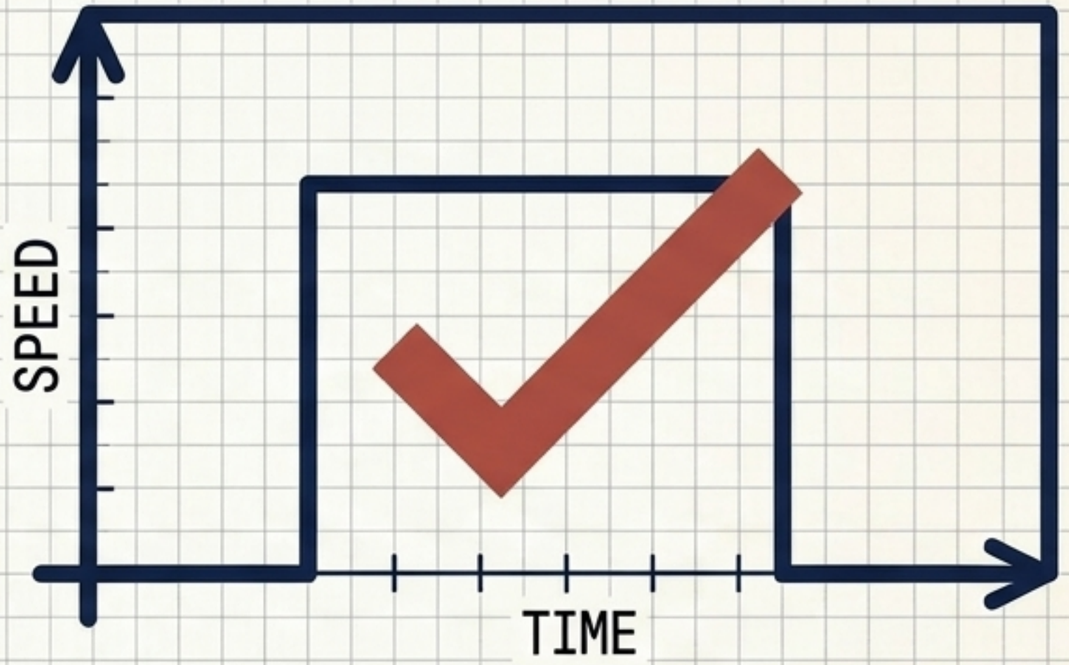
DWG. NO. 042-B

Gemini Notebook

RIGID DYNAMICS: MOMENTUM VS. VELOCITY



Momentum / Friction (Real-World)



Velocity (Digital Arcade)

ZERO MOMENTUM: Objects do not deform; no explicit friction.

DIRECT VELOCITY CONTROL: Players command a state: Input * ConstantVelocity.

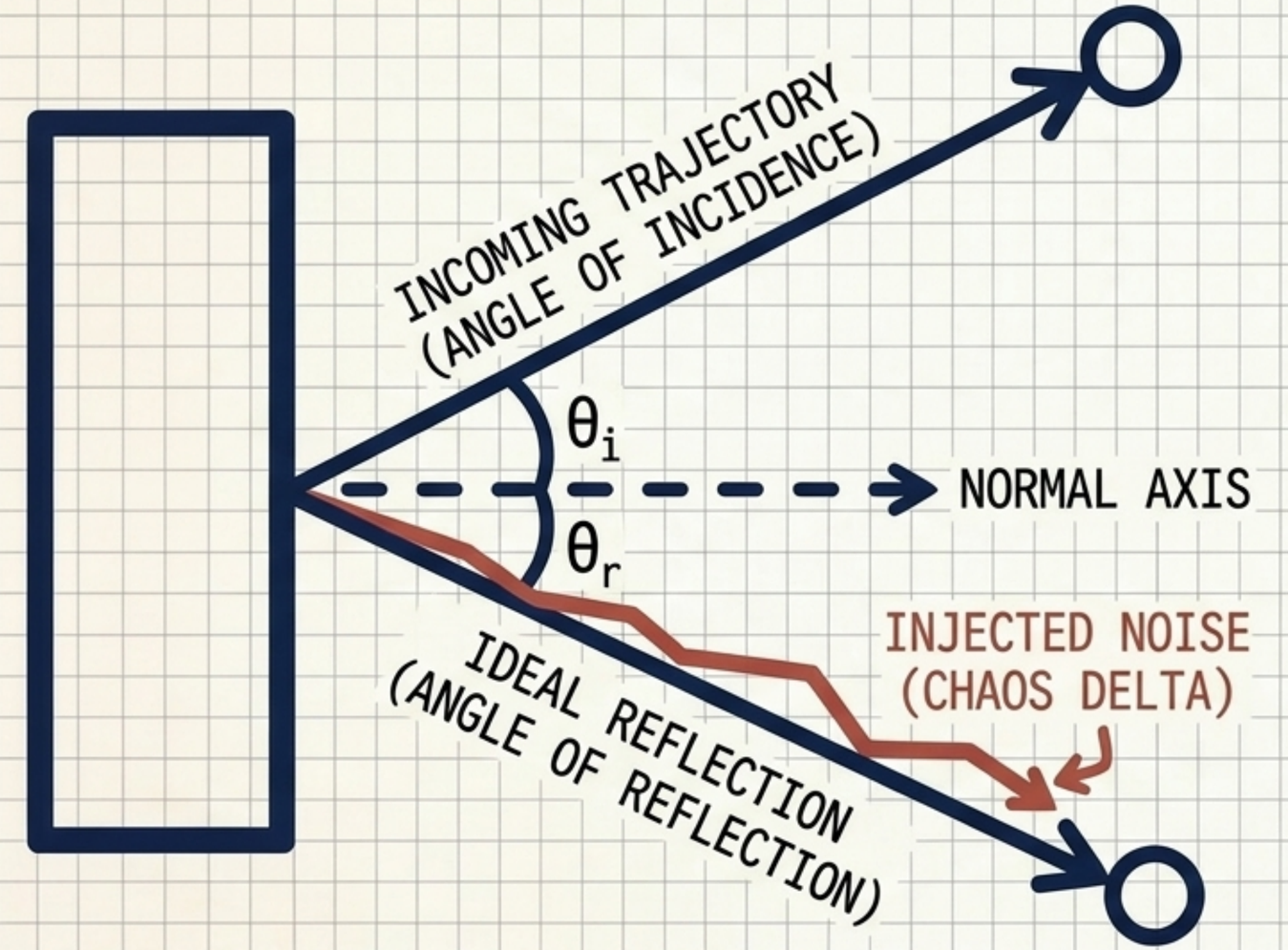
INSTANT STOP: Releasing input drops velocity instantly to zero. No sliding.

DWG. NO. 843-5

Gemini Notebook

PHYSICS: REFLECTION & THE HORIZONTAL TRAP

RIGID DYNAMICS: PERFECT BOUNCE & NOISE INJECTION



LAW OF REFLECTION:
Baseline trajectory mirrors across the normal axis.
$\theta_i = \theta_r$ (Perfect Geometry).

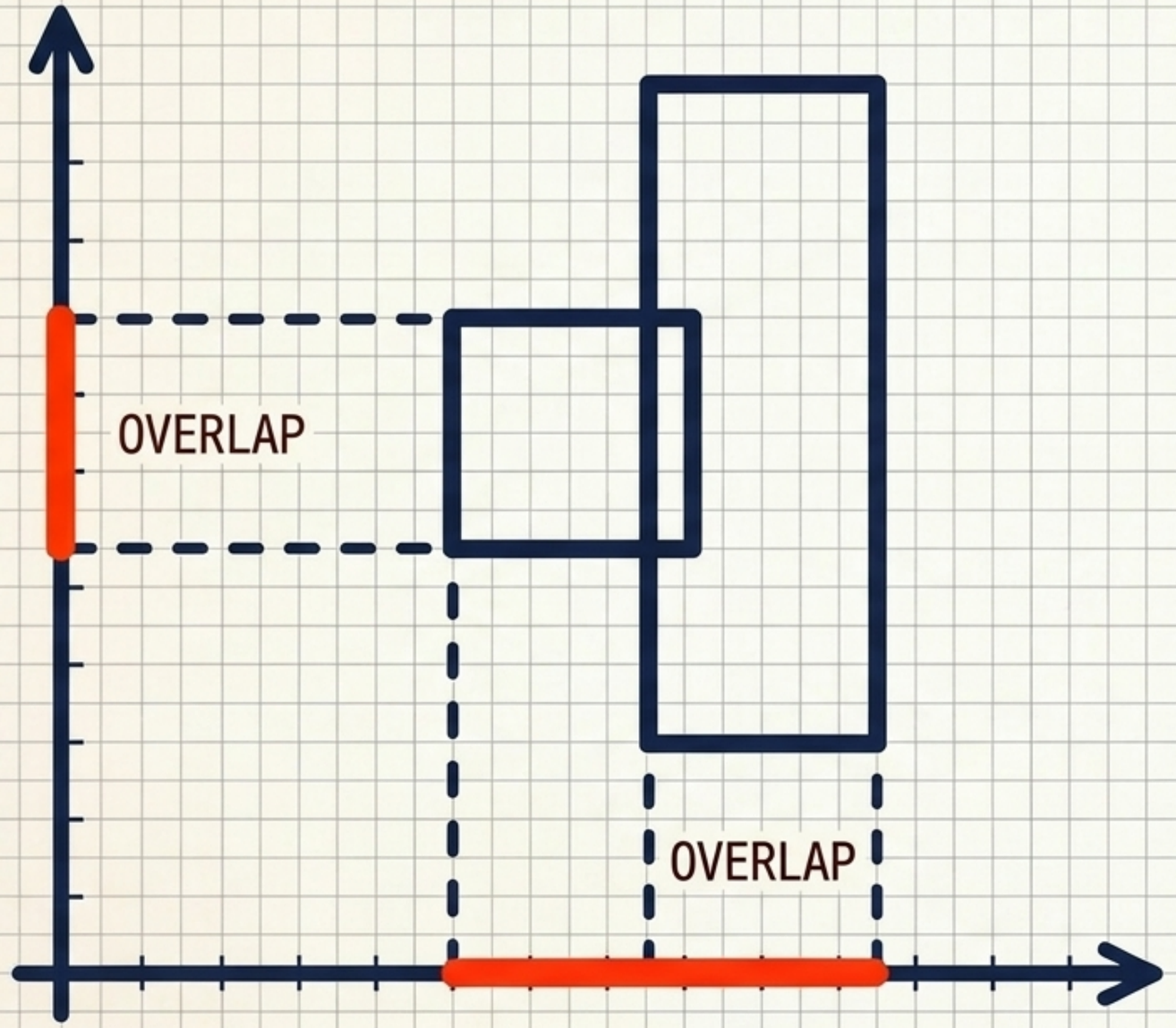
THE HORIZONTAL TRAP:
Pure mathematical reflection causes endless, flat horizontal loops. System Stagnation.

THE SOLUTION:
Inject a tiny delta of chaos (noise) into the reflected velocity to keep the system resolving.
$v_{reflect} = v_{mirror} + random_noise$.

DWG. NO. 843-5

Gemini Notebook

THE OBSERVER: AXIS-ALIGNED BOUNDING BOXES



SEPARATION OF CONCERNS:
A dedicated CollisionEngine handles all intersections.

AABB DETECTION:
Project rectangular bounds onto separate X and Y axes.

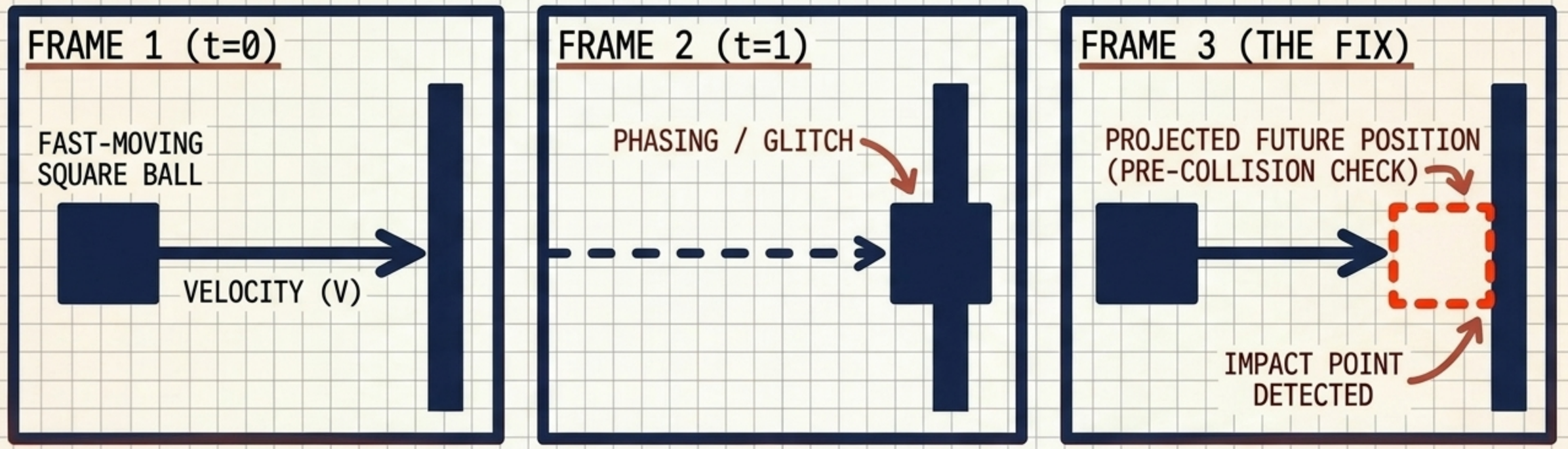
THE RULE OF INTERSECTION:
Collision ONLY occurs if projected intervals intersect on BOTH axes simultaneously.

DWG. NO. 842-C

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TEMPORAL PHYSICS: THE PHASING PROBLEM

DISCRETE TIME STEPS: VELOCITY PROJECTION & COLLISION RESOLUTION



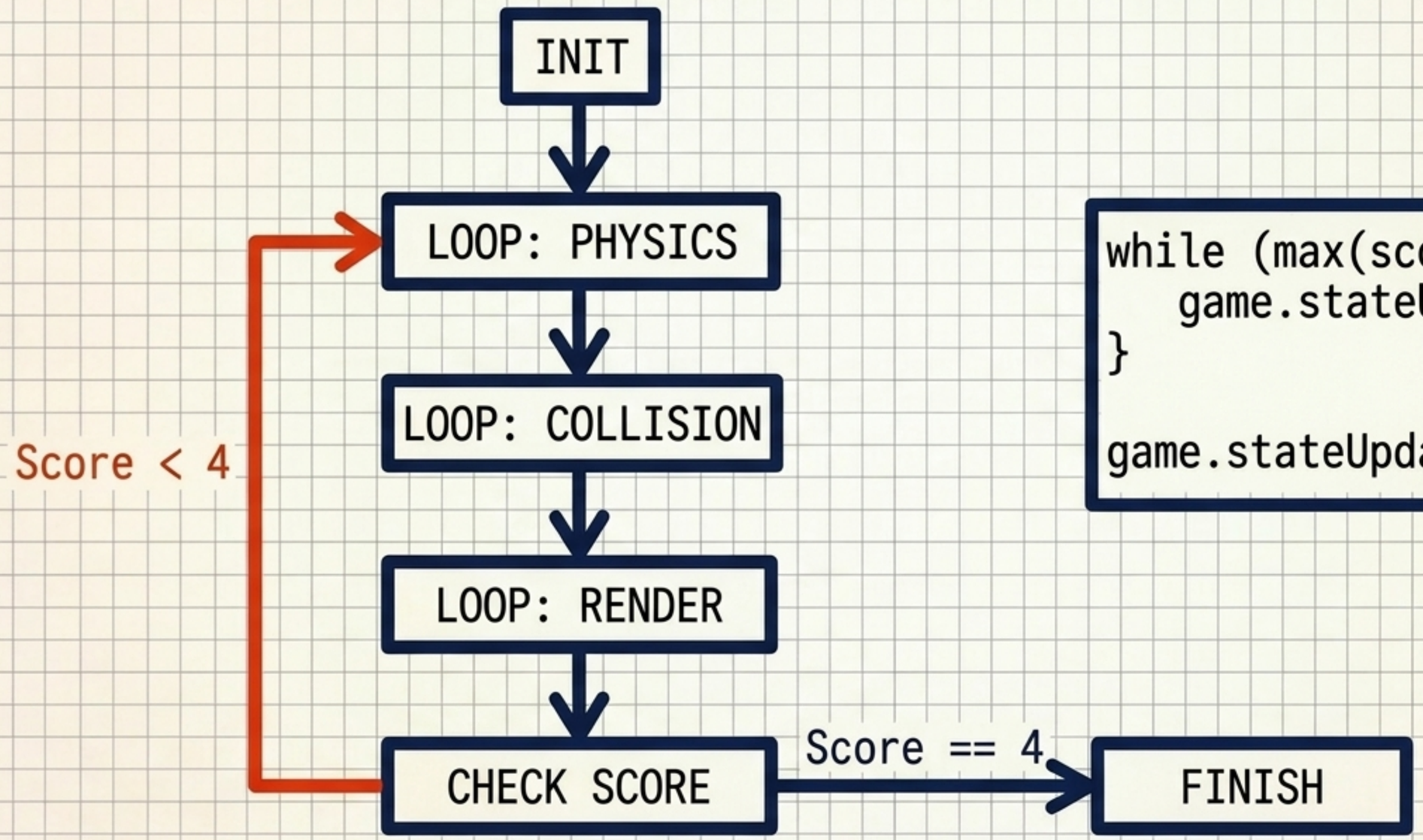
THE ISSUE:
In discrete time steps, fast objects don't slide—they teleport through solid bounds. The displacement between frames exceeds the object's dimension, bypassing the collision check.

THE SOLUTION:
The engine must project future velocity vector to resolve impacts before the teleportation occurs. Continuous collision detection (CCD) or a raycast ahead of the object's motion path is required.

DWG. NO. 843-A

Gemini Notebook

THE HEARTBEAT: GAME LOOP STATE MACHINE




```
while (max(score) < 4) {
    game.stateUpdate(LOOP);
}

game.stateUpdate(FINISH);
```

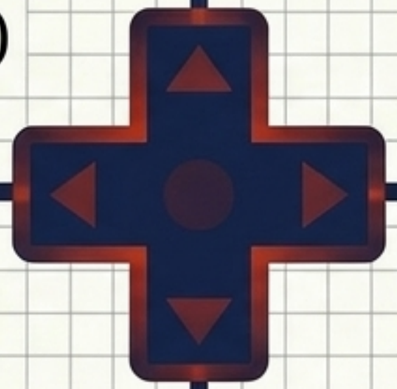
DWG. NO. 843-I

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MINIMALIST INPUT: THE 2-BIT CONTROLLER MAP

00 = NULL
(No input, Velocity = 0)

01 = UP
(Velocity = +y)



10 = DOWN
(Velocity = -y)

11 = SELECT
(Mapped to UI/Menus)

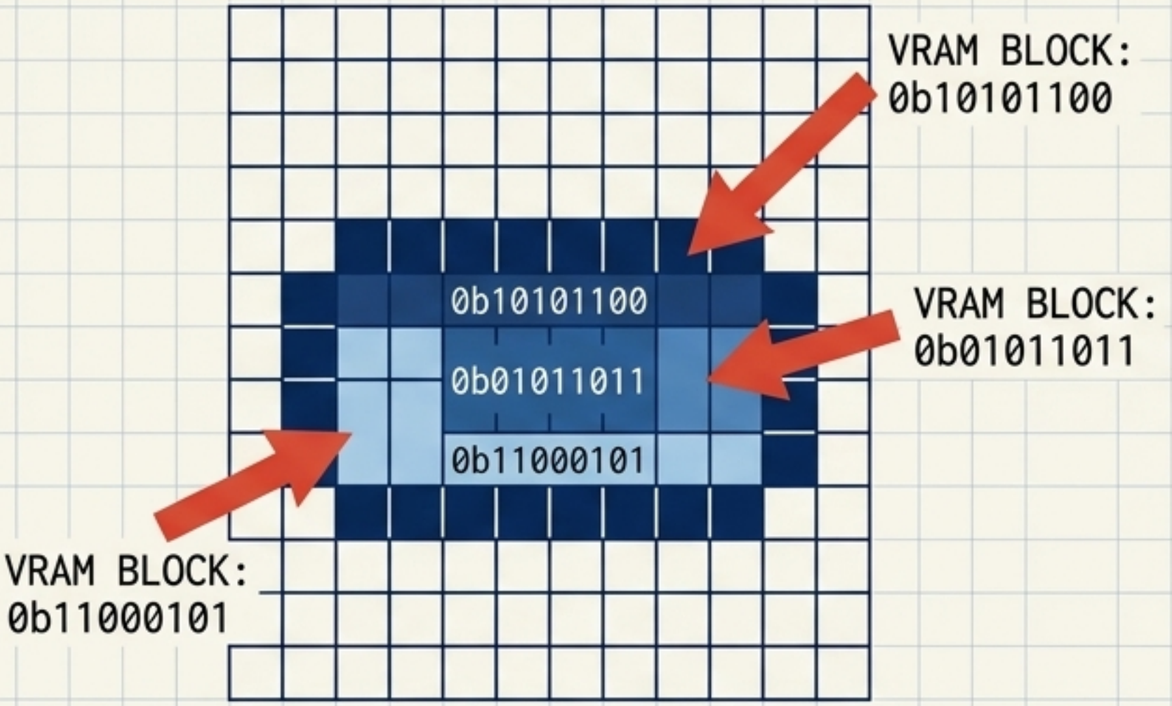
Elegance in system design means minimal data waste. A 2-bit system yields exactly the 4 discrete states required.

DWG. NO. 042-1

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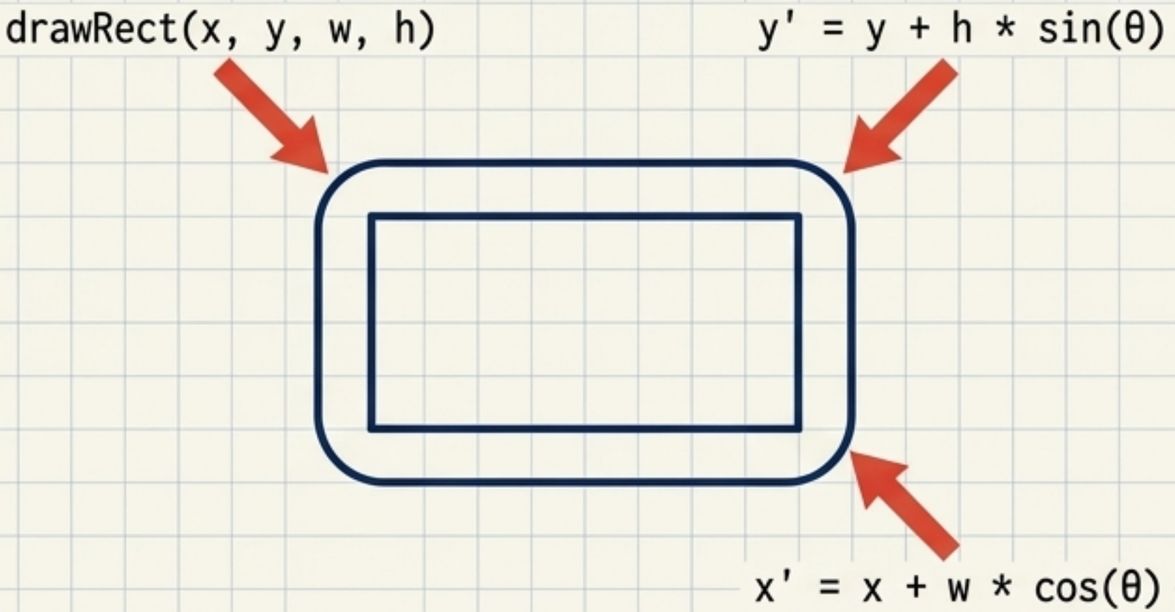
THE SHELL: SPRITES VS. VECTORS

SPRITES



Bitmaps in video memory.
Asset-heavy, highly efficient transformations.

VECTORS

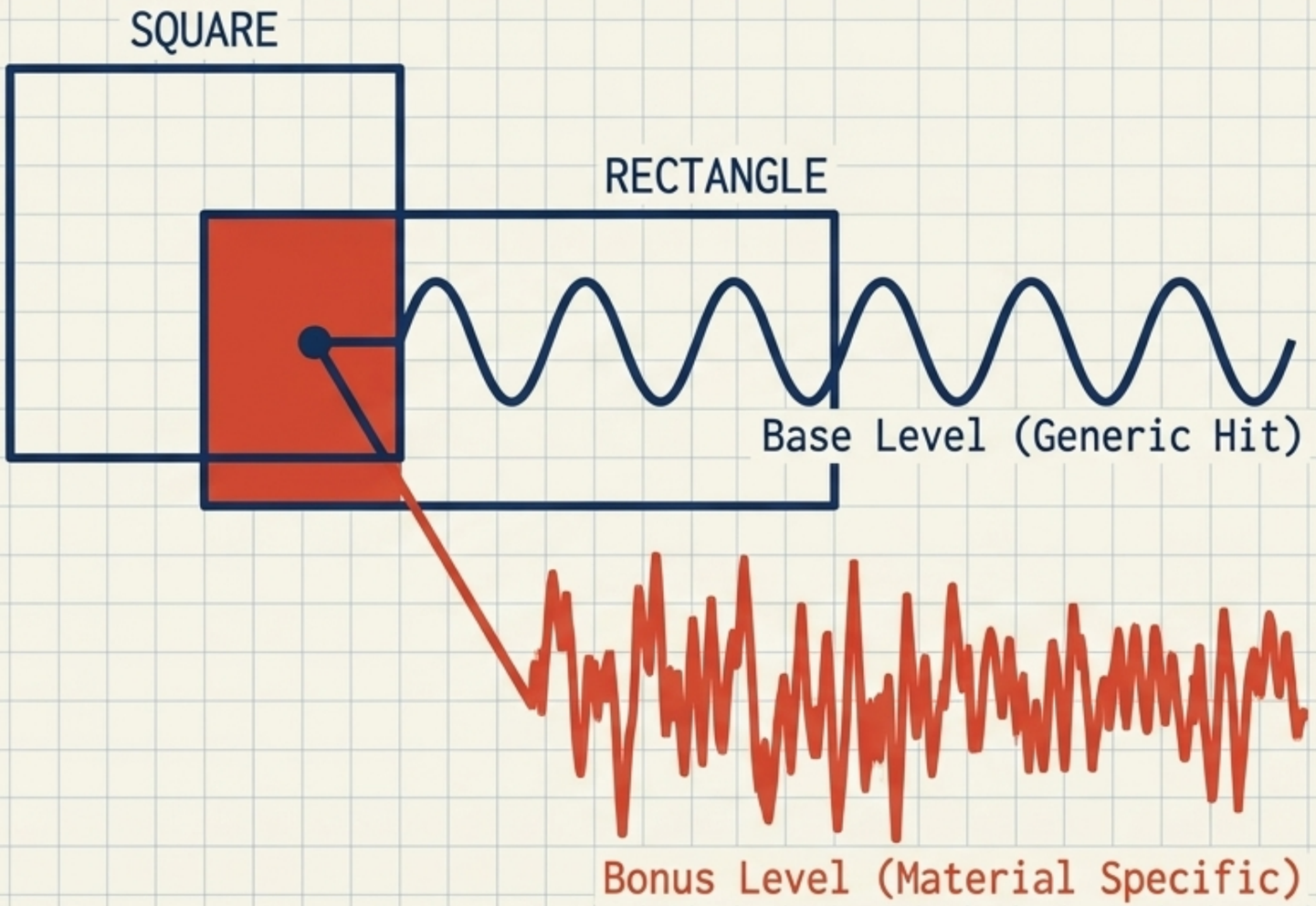


Algorithmically drawn shapes.
Resolution independent, manual fills.

Architectural Win: Because logic is isolated, visual paradigms swap effortlessly without touching physics.

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THE SHELL: AUDIO AS EVENT DATA

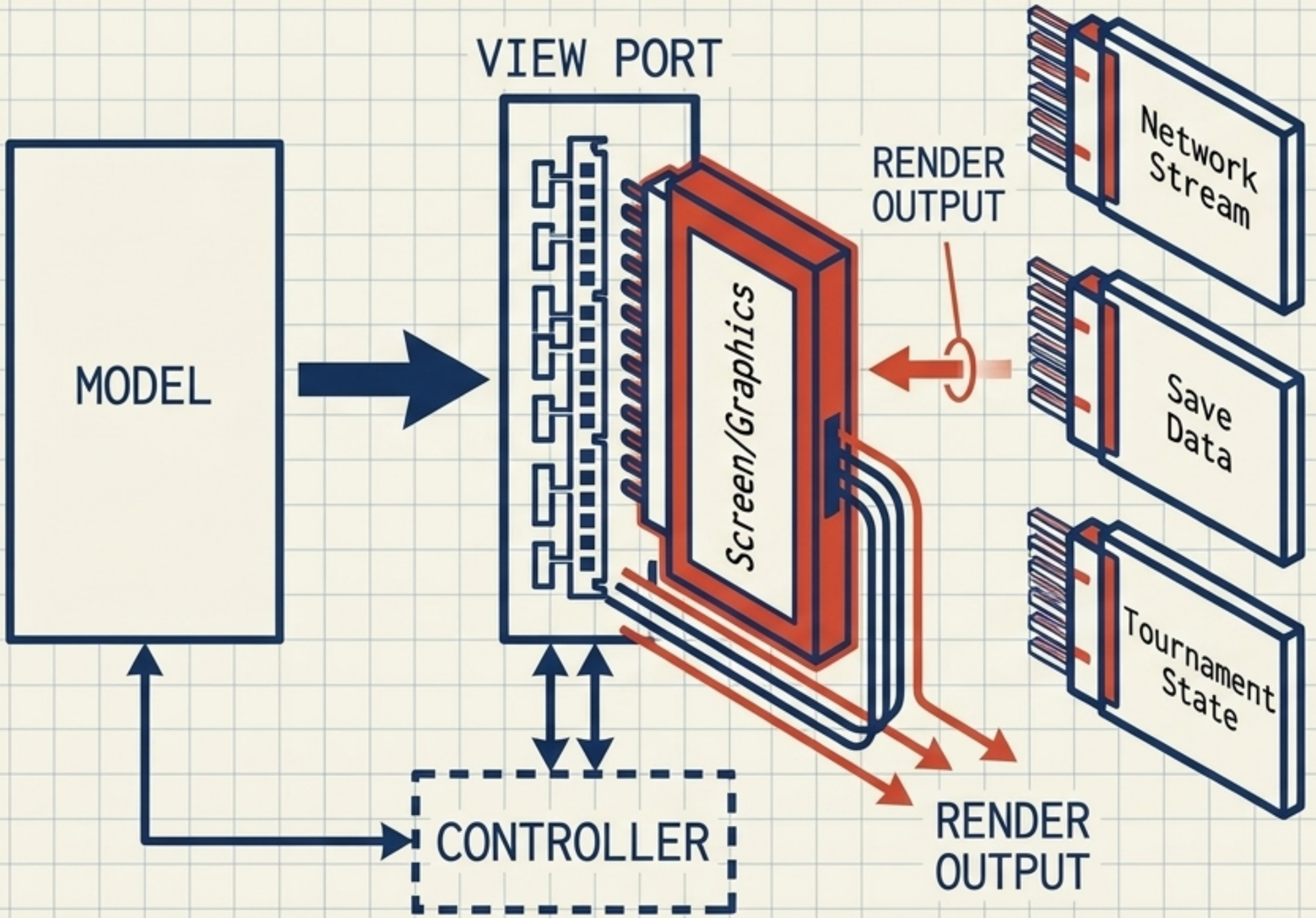


IMPLEMENTATION STRATEGY:

- Audio generation is tied directly into the Collision Engine's event hooks.
- Base Level: Trigger generic acoustic clip on any positive AABB intersection.
- Bonus Level: Unique acoustic signatures for walls vs. paddles, distinct scoring chimes for P1 vs. P2.

Gemini Notebook

SYNTHESIS: BUILDING THE PONG API



- Thinking Bigger: Wrap the game loop in a proper application architecture.
- Expose methods to control, set, and query the game state mathematically.
- With a perfect domain, adding multiplayer networking or stat tracking is just plugging into the existing API.

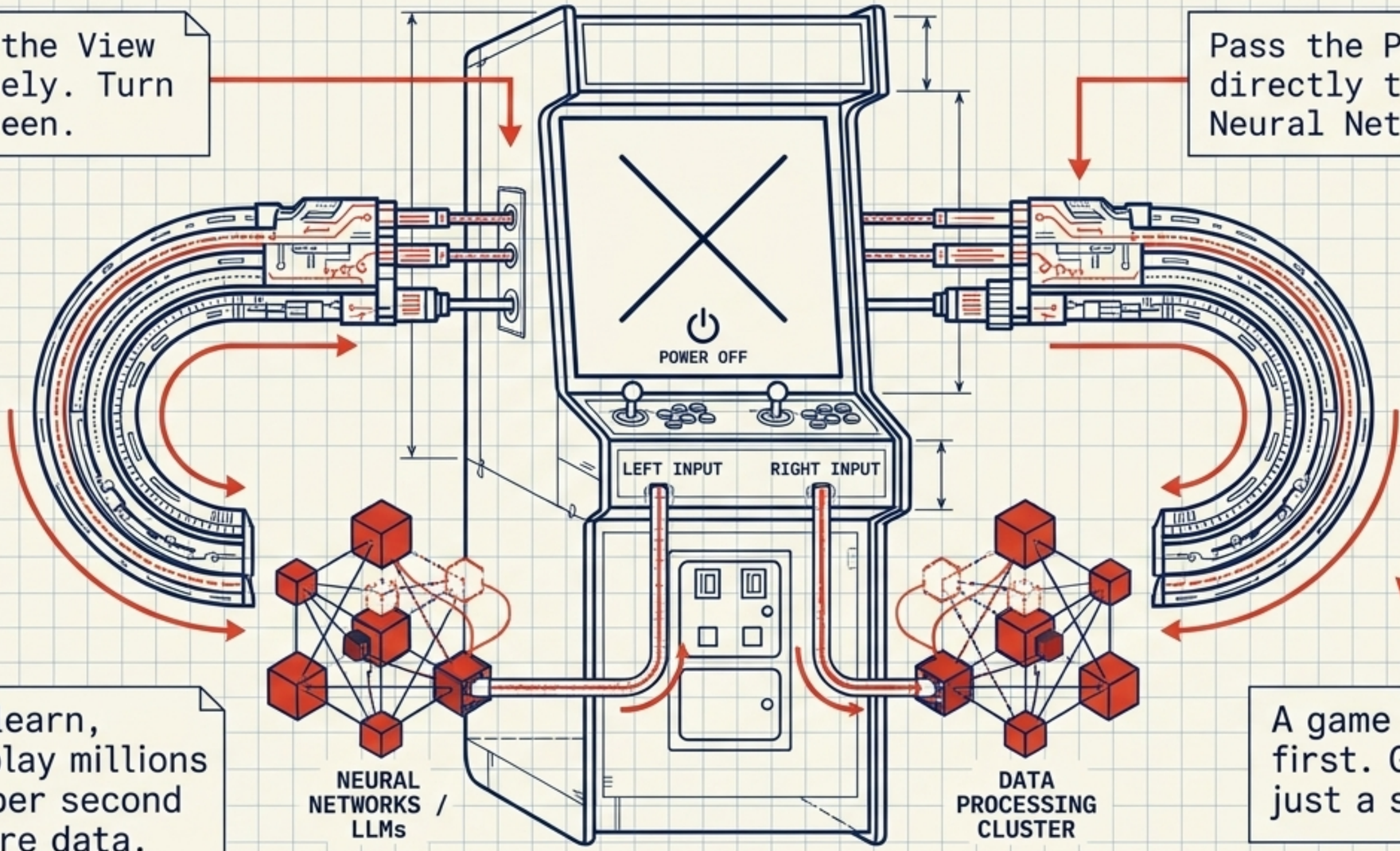
DWG. NO. 042-M

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THE ULTIMATE FLEX: PLAYING IN THE DARK

Disconnect the View layer entirely. Turn off the screen.

Pass the Pong API directly to an LLM or Neural Network.



Watch them learn, react, and play millions of matches per second second in pure data.

A game is a system first. Graphics are just a suggestion.

DWG. NO. 042-N

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